Sahil Karande

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GitHub — LinkedIn — Portfolio

Professional Summary

I build **AI-driven systems** that bring together machine learning, data engineering, and real-time processing. As a **Project Engineer at C-DAC Mumbai**, I work on scalable applications and backend systems while also mentoring students and professionals in AI and data technologies. My projects span computer vision, real-time pipelines, and intelligent automation. I am looking to work on **real-world engineering problems using Python, AWS, Data Engineering, and AI/ML**, with a focus on systems that are practical, scalable, and impactful.

Experience

Project Engineer *Oct 2025 – Present*
Centre for Development of Advanced Computing (C-DAC), Mumbai

- Develop AI-driven applications, backend systems, REST APIs, and scalable data pipelines using Python, databases, and cloud technologies.
- Build solutions involving **AI/ML, computer vision, face recognition, Generative AI, and LLMs**.
- Work with AWS services including EC2, S3, Lambda, **SageMaker, and Amazon Bedrock** for AI and data workloads.
- Deliver technical sessions on Python, Data Science, ML, and Big Data while mentoring students and professionals on end-to-end project development.

IIOT Intern *Dec 2023 – Jun 2024*
Solar Industries India Limited, Nagpur

- Analyzed industrial operational data using Python, Excel, and Power BI to identify production trends and process bottlenecks.
- Automated recurring reporting workflows using Python, reducing manual effort and improving reporting efficiency.
- Developed analytical dashboards and reports to support operational and data-driven decision-making.

Technical Skills

Programming: Python, SQL
Data Engineering: Spark, PySpark, Hadoop, Kafka, ETL, Data Pipelines, Real-Time Processing
AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, OpenCV, Face Recognition
Generative AI: LLMs, RAG, ChromaDB, Gemini, Prompt Engineering
Cloud: AWS EC2, S3, Lambda, SageMaker, Amazon Bedrock, Azure
Backend & Databases: FastAPI, Flask, REST APIs, SQLAlchemy, MySQL, MongoDB, SQLite
Tools: Power BI, Tableau, Streamlit, Excel, Git, GitHub

Selected Projects

NETRA-I – AI-Powered Examination & Proctoring Platform *C-DAC Mumbai*
Python, Flask, SQLAlchemy, Computer Vision, JavaScript, REST APIs

- Developed a secure examination platform with **AI-assisted proctoring**, student verification, PRN validation, OTP verification, and face calibration.
- Implemented examination timers and tab-switch monitoring to track activity during controlled digital assessments.
- Built faculty-side examination management, analytics, authentication, and secure session workflows.

AI-Powered Face Recognition Attendance System *C-DAC Mumbai*
Python, FastAPI, ArcFace, SCRFD, MiniFASNetV2, ONNXRuntime, SQLAlchemy

- Developed a webcam-based automated attendance system using **SCRFD face detection and ArcFace facial embeddings** for real-time staff identification.
- Integrated MiniFASNetV2 anti-spoofing and liveness checks to detect photo, screen-replay, glare, and other spoofing attempts.
- Automated IN/OUT attendance, shifts, overtime and weekly-off calculations, with dashboards, audit logs, absence alerts, and Excel muster exports.

Therapy Connect – AI-Powered Emotion Analysis Platform
Python, Whisper, BiLSTM, RAG, ChromaDB, Gemini, Flask, AWS

- Developed an AI platform combining Whisper speech-to-text with a **BiLSTM emotion-classification model**.
- Built a RAG pipeline using MiniLM embeddings and ChromaDB for context-aware information retrieval.
- Integrated Gemini to generate contextual reflective questions and personalized responses.

Education

PG Diploma in Big Data Analytics *Feb 2025 – Aug 2025*
Centre for Development of Advanced Computing (C-DAC), Mumbai 61.3%

B.Tech in Computer Engineering *Jul 2020 – Jul 2024*
St. Vincent Pallotti College of Engineering & Technology, Nagpur CGPA: 8.13/10

Achievements & Leadership

- **Technical Class Representative – C-DAC PG-DBDA Mumbai (2025)** Represented a batch of 77 students and coordinated academic and technical communication between students and faculty.
- **Research Paper – ICTIA 2024** Presented "Furrify: A Pet Community Website" at ICTIA 2024. Publication
- **Technical Lead – Google Developer Student Clubs (GDSC), SVPCET** *2021 – 2023*
  Led technical initiatives, supported peer learning, and contributed to student-focused technology activities.